

Exercise Set 1

Problem 1

Task a

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv("train.csv")
```

```
df.drop(columns=['id', 'partlybad'], inplace=True)
```

Task b

```
df[['T84.mean', 'UV_A.mean', 'CS.mean']].describe()
```

	T84.mean	UV_A.mean	CS.mean
count	450.000000	450.000000	450.000000
mean	6.439594	10.797330	0.003083
std	9.827282	6.626678	0.002246
min	-23.627710	0.438965	0.000343
25%	-0.733160	4.304971	0.001436
50%	7.661622	11.506937	0.002548
75%	14.118346	16.573188	0.004119
max	27.001804	22.500037	0.013706

Task c

```
arr = df['T84.mean'].values
mean = arr.mean()
std = arr.std()
print(f"Mean: {mean}, Std: {std}")
```

Mean: 6.439594405677285, Std: 9.816356767300478

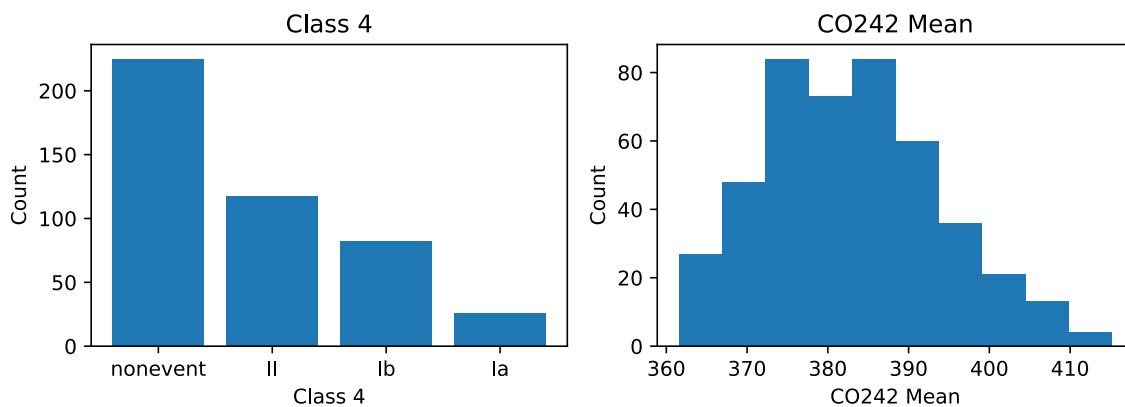
Task d

```
fig, axes = plt.subplots(1, 2, figsize=(8, 3))

axes[0].bar(df['class4'].value_counts().index,
df['class4'].value_counts().values)
axes[0].set_title('Class 4')
axes[0].set_xlabel('Class 4')
axes[0].set_ylabel('Count')

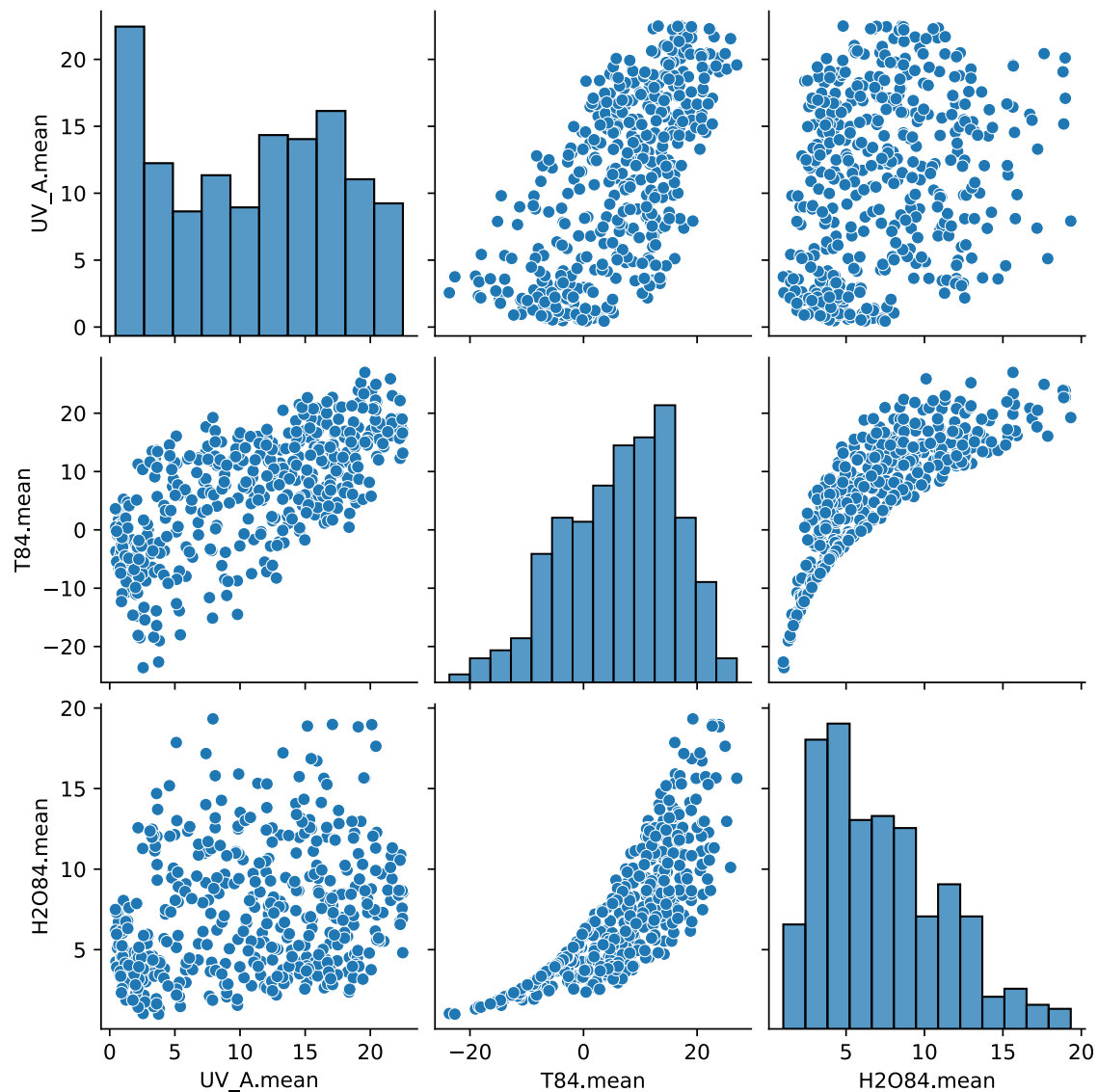
axes[1].hist(df['CO242.mean'])
axes[1].set_title('CO242 Mean')
axes[1].set_xlabel('CO242 Mean')
axes[1].set_ylabel('Count')

plt.tight_layout()
plt.show()
```



Task e

```
sns.pairplot(df[['UV_A.mean', 'T84.mean', 'H2084.mean']])
plt.show()
```



Task f

```
most_common_class = df['class4'].value_counts().idxmax()
p = (len(df) - df['class4'].value_counts()['nonevent']) / len(df)

test = pd.read_csv("test.csv")
submission = pd.DataFrame({'id': test['id'], 'class4': most_common_class, 'p':
p})
submission.to_csv("submission.csv", index=False)
```